

Troubleshooting — symptom first

Every entry here is something that actually happened on this plant, with what it turned out to be. For codes, see `FAULT_CODES.md`.

The drive will not talk

symptom	cause	fix
no reply at all	wrong baud	the baud is per drive (BAUD_COM2); this plant's well drive is 9600 . Tools scan it now and say so
a tool says "no drive ... silent at every baud"	the drive, not the PC	the port opened, so it is not the laptop. Powered? A/B on COM2 , not COM1? Polarity?
no reply, port opens	the dashboard is running	it holds the port — close that window
Access is denied on the port	another tool still has it	close it; a killed shell can leave its Python child holding the port
port vanished entirely	the drive is unpowered	on a PV build the adapter dies with the bus. Check the array
reads work, numbers are nonsense	the PC map does not match the firmware	run <code>check_link_api.py</code> ; re-run <code>UPDATE.bat</code> to refresh tools/

It runs, but not as well as it should

symptom	cause	fix
frequency sits well below FMAX in good sun	the slip cap fed by a noisy rotor estimate	the rotor estimate swings ± 3 Hz; if the cap's input is lightly filtered its noise minima clamp the setpoint. Fixed in firmware by filtering harder — make sure the drive has a current image
power seems capped at a round number and will not rise	the DC current sensor is railing	run <code>CHECK_SENSOR.bat</code> . See <code>MANUAL.md</code> — the offset pot
tracker parks low and will not climb	wrong MPPT margin	F12 is the margin $\times 10$. A stale EEPROM value of 35 means margin 3.5, which parks far short
climbs, collapses, climbs again, about once a minute	the margin is too aggressive for that irradiance	the MPP of a constant-power load is a cliff , not a hill — overshoot does not cost a little power, it costs the run. Raise F12
water stops at dusk but no fault	LINGER (13)	correct behaviour — it was below pump cut-in and stopped fighting

A useful cross-check: compare `F0C_PEST` against $VDC \times IDC$. Motor power cannot exceed DC input power. If the ratio jumps above ~ 1.0 , the DC current reading is wrong — that is what exposed the railed sensor.

Starting

symptom	cause	fix
pre-start fails right after a stop	the motor is still coasting; back-EMF looks like a fault	wait ~ 2 s and start again
pre-start reports an open phase that is not open	relay or ADC-offset chain	cost a day on the grid board, 2026-09-19. Check the relay actually pulled in, and the offsets
asymmetry (31) that makes no sense	a probe vector aborted on over-current, leaving no measurement	check <code>PS_FLAGS</code> for <code>0x4000</code> first
ten UV trips in a row at dawn	not enough array to magnetise	expected on a weak morning. <code>START_DEFER</code> (12) now handles it

symptom	cause	fix
drive reads 25 V on a live 510 V bus and sits in FAULT_UV_WAIT	wrong V_DC_SENSED_OFFSET for that board variant	wrong image for the board. Check which variant it is

Tuning

symptom	cause
fault 44, rotor did not turn	check the plate first. The gate is 566 / plate_I_rms mH, so a plate current entered *too low raises* the gate
tune fails only on PV, passes on a stiff supply	not enough watts. Tune on the stiffest supply available
tune passes but disagrees with the plate (47)	it may not be the motor you think it is
sigma-Ls looks too high	dead time. It stays at the shared 188 — under-compensating inflates apparent sigma-Ls and turns a marginal tune into a reliable 44

Flashing

symptom	cause	fix
OTA stalls part way	an old bootloader (CAB6EC2A) cannot be repaired over serial	SWD recovery. Check which bootloader the drive runs before blaming the cable
every app "dies at boot" after a bootloader flash	a Debug bootloader was flashed — 4260 B into a 4096 B slot, overrunning the app region	flash the -0s Release bootloader, 3268 B, from boot loader /
new app refuses to boot after SWD	stale OTA metadata: it carries the *old* image's length and CRC	mass erase first; leave 0x0800F800 erased
random OTA corruption over RS485	an auto-direction adapter clipping the last byte	fixed in bootloader 18274D72 with a pad byte, reply lead and per-chunk CRC. Check the bootloader version
FLASH.bat refuses the board	no STABLE image for it — nobody has proven one	ask the engineer. Do not flash LATEST on your own judgement

Settings that will not stick

The usual cause is no SAVE_SETTINGS — a write is RAM only and dies at the next reset, and these boards reset often. One board showed 104 resets.

It looks exactly like the drive "reverting" a good write. It is not. Save, then RELOAD_EEPROM, then read back.

Things that look like faults and are not

- STOP_REASON **11** — somebody sent a stop over the link. Including a tool.
- STOP_REASON **12 / 13** — deliberate probes, not failures.
- BUS_SHED dipping on a cloud edge — the droop doing its job.
- FOC_TUNE_STS reading 33 — it is **packed**, not a plain flag: $\text{tuned} + 2 \times \text{fail} + 4 \times (\text{state} \& 0xF) + 64 \times (\text{failcode} \& 0xF)$. A healthy tune parked at the last stage reads $1 + 4 \times 8 = 33$.
- MEAS_* registers reading 0 while TEMP, VDC, IDC are live — that mirror group is unpopulated. Cosmetic.
- MOD_IDX reading a flat 0.2000 — also unpopulated. Use FOC_VDEM against $V_{dc}/\sqrt{3}$ instead.

When to stop and ask

- OC_INST (23) — likely hardware, not load.
- INSUL_LEAK (30) — megger the motor and cable **before** restarting. There is a wet joint 70 m down until proven otherwise.
- An OTA that stalled part way.
- Anything where the fix would mean changing a protection limit.